forward: source → field (unique)

inverse: wanted field → source (not unique)

1 target and limits

ROI field must obey Maxwell:
write it in a harmonic basis
shielding: exterior field ≈ 0
limits: power, voltage, force, pitch

2 source space

former surface (plate, cylinder)
$\psi_s = c_1 N_1 + c_2 N_2 + \cdots$
current continuity is built in
ψ_s constant on edges: no leakage

3 consistent operators

field: $\mathbf{b} = \mathbf{C}\,\mathbf{c}$
power: $P = \mathbf{c}^T \mathbf{R}\,\mathbf{c}$
energy: $W = \frac{1}{2}\mathbf{c}^T \mathbf{L}\,\mathbf{c}$
one basis, one set of units

4 solve, trade off

regularise: error + α·P + β·W
or: exact constraints, least power
sweep α, draw the trade-off
singular values: costly modes

5 extract the winding

contours at $\Delta\psi_s = I_w$
leads, series links, crossovers
check pitch, width, bend radius
PCB: copper weight, vias

6 forward re-check

recompute field, R, L of the wires
eddy currents nearby (Ch. 12)
forces (Ch. 15), PNS (Ch. 21)
compare with measurement

fails: back to steps 1–2, change target, source space or limits